XMUM Library Walk-In and Gate-Count Analysis

Verified Patterns, Scan Events, and Data Quality — 2023–2026

MAT Library Data Talk

9 October 2026

Source: provided XMUM Library walk-in dataset.

First: What Are We Counting?

2023–2025: Gate counts

Four daily gate-meter readings (L1 A, B, C; L2 D). They count gate activity, **not distinct visitors**.

- 2023 begins on 26 May.
- 2024 has near-full daily coverage.
- 2025 has gaps and repaired totals.

2026: ID scan events

Timestamped IN and OUT scans from many workbook snapshots. We de-duplicate across files.

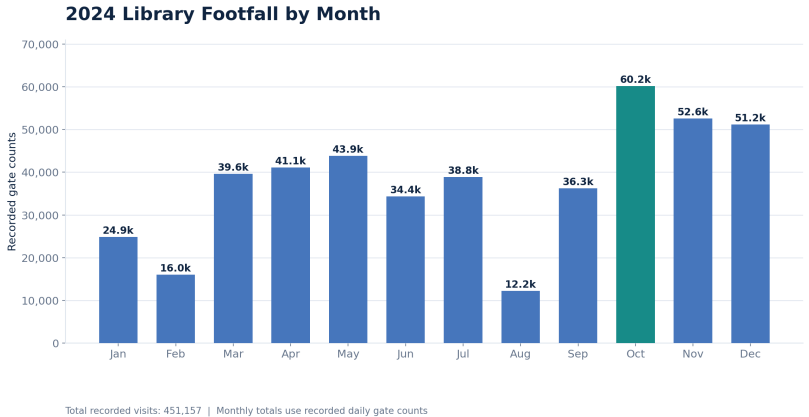
- IN/OUT *events* are not unique people.
- Scan coverage is irregular across months.
- **Do not merge** these with gate counts.

Executive Summary

- **2024: 451,157** recorded gate counts across 363 usable days.
- **October 2024: 60,180** counts, the highest monthly total in 2024.
- **L2 Gate D: 70.5%** of 2024 gate readings.
- **2026: 340,206** unique scan events after de-duplicating 2,651,519 event-row occurrences.

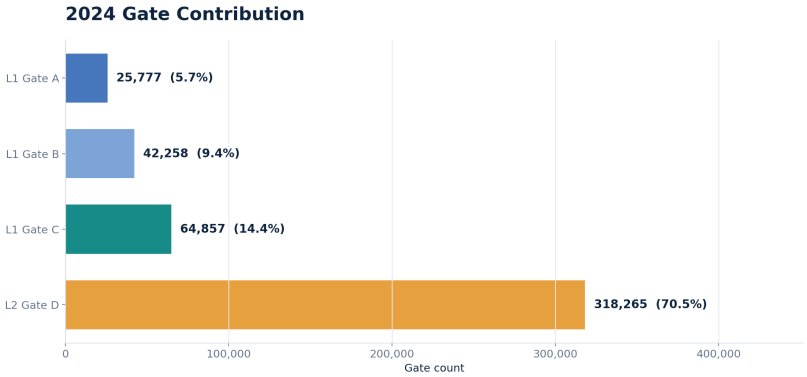
The gate and scan datasets use different measurement systems; there is no defensible direct year-on-year comparison.

Monthly Footfall: 2024



2024 offers the strongest single-year basis for describing monthly demand.

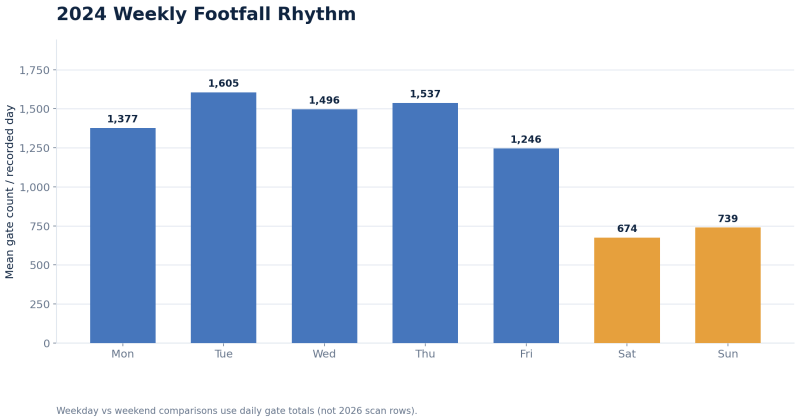
Where Is the Gate Traffic?



2024: four gate series, summed over days with recorded gate values. Counts reflect gate readings, not unique persons.

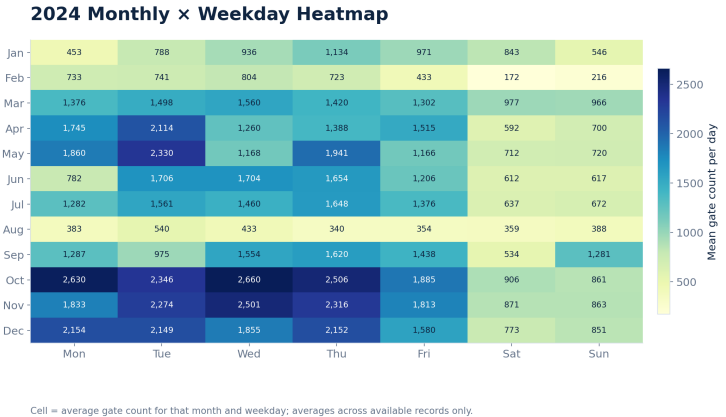
L2 Gate D is the dominant meter; investigate whether its role or location affects counts.

Which Days Are Busy?



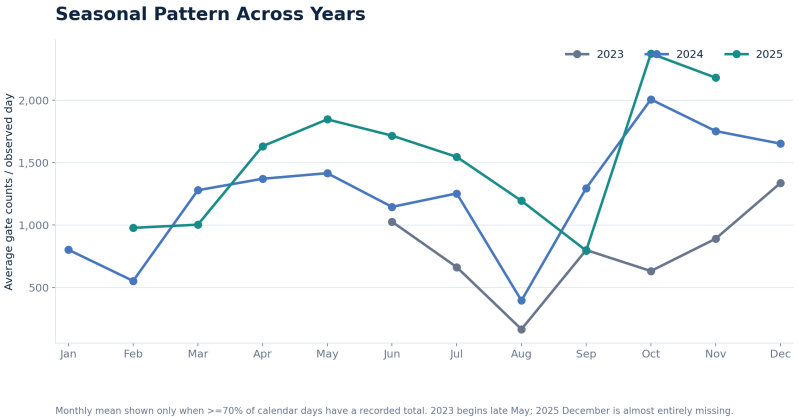
The mean recorded gate count is 1,452 on weekdays versus 707 on weekends in 2024.

Recurring Patterns in 2024



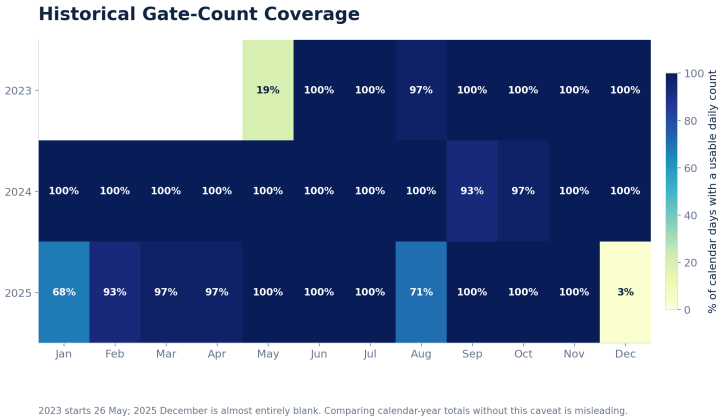
The matrix averages observed days, rather than counting missing observations as zeros.

Across Years: Compare with Caution



Only months with at least 70% daily coverage are plotted in this average-per-day comparison.

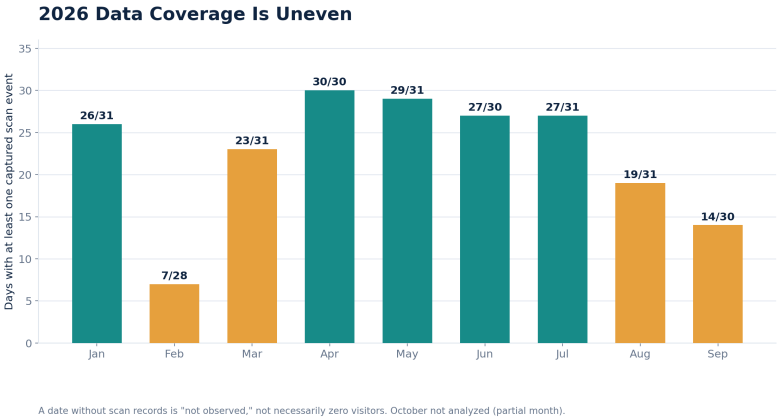
Coverage Audit of Historical Data



2023 starts 26 May; 2025 December is almost entirely blank. Comparing calendar-year totals without this caveat is misleading.

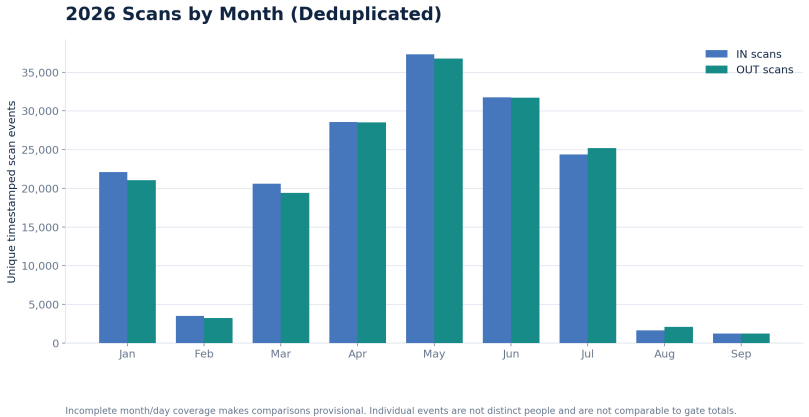
2023 begins late May and December 2025 has effectively no recorded gate counts.

2026 Scan Coverage Is Incomplete



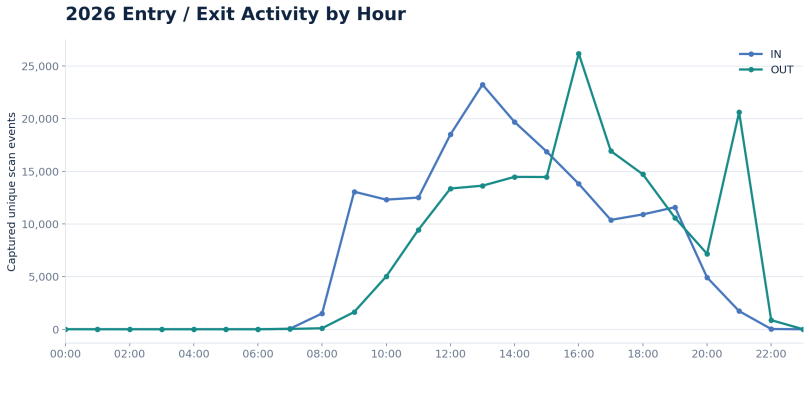
January–September coverage varies greatly; missing days must not be labelled zero visits.

2026 Monthly Scans, After De-duplication



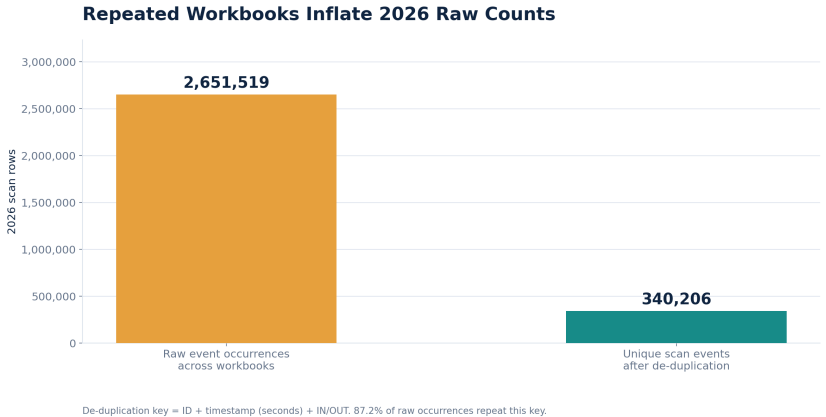
This is the number of timestamped IN and OUT events, not the number of distinct visitors.

2026: Activity by Hour



IN scanning peaks around 13:00 and OUT scanning around 16:00 in the available records.

Why De-duplication Matters



About 87.2% of raw 2026 event occurrences repeat an ID + timestamp + direction already seen.

Important Data Quality Corrections

1. **Repeated snapshots:** scans were de-duplicated globally across source workbooks.
2. **Inconsistent IN/OUT headers:** 16 workbook sheets contradict their sheet name; classification follows the actual header.
3. **2023 dates:** early Nov/Dec dates require month-label and row-order correction.
4. **2025 totals:** 44 missing daily totals recovered from four recorded gates; two reported totals disagree with gate components.
5. **Missingness:** blank values were not treated as zero; anomalous spikes should be verified against source equipment.

No student IDs are disclosed in the charts or in the aggregated scan tables.

Implications and Next Steps

What the evidence supports

2024 demand changes markedly by month and weekday; the L2 gate is central to recorded gate activity. 2026 scan timings offer operational clues, not an occupancy estimate.

Recommended follow-up

Standardize daily exports; keep immutable source logs; reconcile IN/OUT and gate counters; investigate outlier days; publish coverage and validation rates alongside every trend.

Conclusion: analyze source quality before interpreting visitor trends.